

TECHNICAL BLUEPRINT · PROBLEM 2/50

# Forearm *Moment* Solver

*An interactive diagram-driven solver for the forearm moment problem  
from Meriam & Kraige — Engineering Mechanics: Statics*

REFERENCE

Meriam & Kraige, Problem 2/50

TECH STACK

HTML5 · CSS3 · JavaScript · Canvas

DATE

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§ 1 PROBLEM STATEMENT

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**Meriam & Kraige Problem 2/50** — A forearm of mass  $m_{\text{arm}} = 2.3 \text{ kg}$  is held in position by the biceps muscle, which exerts a tension  $T$  at a perpendicular distance  $d_T$  from the elbow pivot  $O$ . The forearm's centre of gravity is at point  $G$ , located  $d_G$  along the arm from  $O$ . A sphere of mass  $m_s$  is held at the hand at point  $A$ , a horizontal distance  $d_A$  from  $O$ .

The student must:

1. Determine the combined moment of the weights of the forearm and sphere about the elbow pivot  $O$ .
2. Find the biceps tension  $T$  required for rotational equilibrium ( $\Sigma M_O = 0$ ).

CORE CONCEPT

This problem is a classic application of the **scalar moment equation**  $M = F \times d$ . The forearm acts as a lever arm, and the biceps must produce a counter-clockwise moment to balance the clockwise moments of the two weights about O.

§ 2 GIVEN DATA & VARIABLES

| Symbol           | Description                                     | Default Value | Unit             |
|------------------|-------------------------------------------------|---------------|------------------|
| $\theta$         | Elbow angle (upper arm to forearm)              | 55            | degrees          |
| $d_T$            | Horizontal distance from O to biceps attachment | 50            | mm               |
| $d_G$            | Distance along arm from O to CG of forearm      | 150           | mm               |
| $d_A$            | Horizontal distance from O to sphere (hand)     | 325           | mm               |
| $m_{\text{arm}}$ | Mass of forearm                                 | 2.3           | kg               |
| $m_s$            | Mass of sphere                                  | 3.6           | kg               |
| $g$              | Gravitational acceleration                      | 9.81          | m/s <sup>2</sup> |

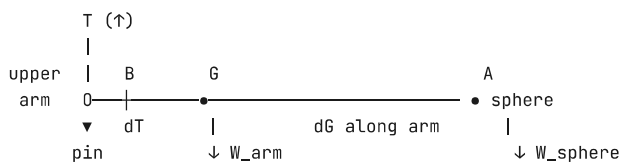
### § 3 PHYSICS & EQUATIONS

#### Free-Body Diagram Description

The forearm is modelled as a rigid bar pinned at **O** (the elbow joint). Three forces act on it:

- **T** — Biceps tension, acting vertically upward at the attachment point B, a horizontal distance  $d_T$  from O.
- **W<sub>arm</sub>** — Weight of the forearm, acting vertically downward at the CG point G.
- **W<sub>sphere</sub>** — Weight of the sphere, acting vertically downward at point A (hand).

The elbow joint provides pin reactions ( $O_x$ ,  $O_y$ ) but these do not create moments about O since their moment arm is zero.



#### Weight Calculations

$$W_{arm} = m_{arm} \times g \quad W_{sphere} = m_s \times g$$

#### Horizontal Moment Arms

Since moments about O use horizontal (perpendicular) distances for vertically-acting forces, the horizontal distance from O to G depends on the arm geometry:

$$x_G = d_G \times \cos(90^\circ - \theta) = d_G \times \sin \theta$$

#### Moment Equation (about O)

Taking counter-clockwise as positive:

$$\Sigma M_O = T \times d_T - W_{arm} \times x_G - W_{sphere} \times d_A = 0$$

### Step 1: Combined Moment of Weights

$$M_{weights} = W_{arm} \times x_G + W_{sphere} \times d_A \quad (\text{clockwise about } O)$$

### Step 2: Required Biceps Tension

$$T = M_{weights} / d_T = (W_{arm} \times x_G + W_{sphere} \times d_A) / d_T$$

#### KEY INSIGHT

Because the biceps attachment is very close to the elbow joint ( $d_T \approx 50$  mm) compared to the distances to the weights ( $d_G \approx 150$  mm,  $d_A \approx 325$  mm), the required biceps tension  $T$  is **much larger** than the combined weight — a classic mechanical disadvantage of the human forearm lever system.

§ 4    **SAMPLE CALCULATION (DEFAULT VALUES)**

Step 1 — Compute weights

$W_{\text{arm}} = 2.3 \times 9.81 = 22.563 \text{ N}$   
 $W_{\text{sphere}} = 3.6 \times 9.81 = 35.316 \text{ N}$

Step 2 — Horizontal distance 0 to G

$a = 90^\circ - 55^\circ = 35^\circ$  (angle of forearm below horizontal)  
 $x_G = 150 \times \cos(35^\circ) = 150 \times 0.8192 = 122.87 \text{ mm}$

Step 3 — Combined moment of weights about 0 (CW)

$M_{\text{weights}} = W_{\text{arm}} \times x_G + W_{\text{sphere}} \times d_A$   
 $= 22.563 \times 122.87 + 35.316 \times 325$   
 $= 2,772.04 + 11,477.70$   
 $= 14,249.74 \text{ N}\cdot\text{mm}$

Step 4 — Required biceps tension for equilibrium

$T \times d_T = M_{\text{weights}}$   
 $T = 14,249.74 / 50$   
 $T \approx 285.0 \text{ N}$

Combined Moment (CW) : 14,249.74 N·mm  
Biceps Tension T : 285.0 N

§ 5    **APPLICATION USER FLOW**

The interactive solver follows a streamlined five-step workflow:



- **Load Page:** The solver loads with default values ( $\theta = 55^\circ$ ,  $m_{\text{arm}} = 2.3 \text{ kg}$ , etc.) and draws the initial diagram.
- **Enter Inputs:** Seven input fields in the right panel allow the student to modify geometry, masses, and gravity. Supports both numeric and symbolic (letter) inputs.
- **Diagram Updates:** As values are typed, the HTML5 Canvas redraws in real time — the forearm rotates to match  $\theta$ , force arrows scale, and dimension labels update.

- **Compute:** Clicking the gold "Compute Solution" button runs the solver.
- **View Solution:** A results panel slides in with two answer boxes (combined moment and biceps tension  $T$ ) followed by detailed step-by-step working.

#### SYMBOLIC INPUT SUPPORT

If the student types a non-numeric value (e.g., "m" for mass), the app displays the symbol on the diagram and in the solution steps, while using the default numeric value internally for calculations and rendering.

§ 6 UI DESIGN & LAYOUT

Screen Layout

The application uses a two-column layout (responsive, stacking on mobile):

- **Left panel (≈70%):** A full-height HTML5 `<canvas>` element showing the anatomically-styled forearm diagram with labelled force arrows, dimension lines, and angle arcs.
- **Right panel (≈30%):** Input fields grouped by category (Geometry / Masses / Constants) with a prominent "Compute Solution" button at the bottom.

Canvas Diagram Elements

| Element          | Visual Style                                                      |
|------------------|-------------------------------------------------------------------|
| Upper arm        | Anatomical gradient fill with biceps/triceps muscle bulges        |
| Forearm          | Gradient-filled tapered body, narrowing towards wrist             |
| Elbow joint (O)  | Radial gradient ellipse + blue pivot dot                          |
| Hand at A        | Simplified hand shape gripping the sphere                         |
| Sphere           | 3D radial gradient with specular highlight                        |
| Biceps tension T | Cyan (#00e5ff) upward arrow at point B                            |
| Weight arrows    | Red (#ff2255) downward arrows at G and A, labelled "m×g N"        |
| Angle arc        | Pink dashed arc from vertical to forearm, labelled $\theta^\circ$ |
| Dimension lines  | Green (#39ff14) lines with arrowheads, labelled in mm             |
| Grid             | Subtle blue grid lines on dark (#050912) background               |
| Legend           | Top-right corner showing force color key                          |

Input Fields

| Field            | Label                                 | Default | Live Update                |
|------------------|---------------------------------------|---------|----------------------------|
| $\theta$         | Elbow angle: upper arm to forearm (°) | 55      | Yes — rotates forearm      |
| $d_T$            | Horiz. distance O to biceps (mm)      | 50      | Yes — moves T arrow        |
| $d_G$            | Distance along arm O to G (mm)        | 150     | Yes — moves G marker       |
| $d_A$            | Horiz. distance O to sphere A (mm)    | 325     | Yes — moves hand/sphere    |
| $m_{\text{arm}}$ | Forearm mass (kg)                     | 2.3     | Yes — updates arrow label  |
| $m_s$            | Sphere mass (kg)                      | 3.6     | Yes — updates arrow label  |
| $g$              | Gravity (m/s <sup>2</sup> )           | 9.81    | Yes — updates arrow labels |



§ 7 COLOUR & DESIGN SYSTEM

| Token                  | Value   | Usage                                 |
|------------------------|---------|---------------------------------------|
| <code>--bg</code>      | #0d0d10 | Page background (deep dark)           |
| <code>--surface</code> | #16161e | Card / panel backgrounds              |
| <code>--gold</code>    | #c8a050 | CTA button, highlights, answer values |
| <code>--blue</code>    | #4a90d9 | Pivot dot, numeric values in steps    |
| <code>--green</code>   | #4dc87a | Live dot indicator, result checkmark  |
| <code>--red</code>     | #e06060 | Reserved for error states             |
| <code>--text</code>    | #f0ead8 | Primary text (warm white)             |
| <code>--muted</code>   | #7a7468 | Labels, secondary text                |

§ 8 TECHNICAL ARCHITECTURE

File Structure

```
mechanics_app/  
└─ problem_2_50.html  ←!— Single-file app: HTML + CSS + JS →  
    └─ <style>         ←!— All CSS: layout, colors, typography →  
        └─ <body>       ←!— Topbar, hero, grid, input panel, result →  
            └─ <script>  ←!— parseField, canvas draw, solver, UI →
```

Key JavaScript Functions

| Function                             | Purpose                                                                      |
|--------------------------------------|------------------------------------------------------------------------------|
| <code>parseField(id, default)</code> | Reads input; returns {val, sym, isNum} — handles symbolic/numeric gracefully |
| <code>drawDiagram()</code>           | Clears canvas and redraws the entire anatomical diagram with force arrows    |
| <code>drawUpperArm()</code>          | Renders humerus with muscle gradients (biceps + triceps bulges)              |
| <code>drawForearm()</code>           | Renders forearm body with tapered Bézier curve and muscle details            |
| <code>drawHand()</code>              | Simplified hand shape at point A                                             |
| <code>drawSphere()</code>            | 3D sphere with radial gradient and specular highlight                        |
| <code>drawArrow()</code>             | Generic force arrow with filled head and offset label                        |
| <code>compute()</code>               | Runs the solver, populates answer boxes and step-by-step working             |

### Auto-Scaling Algorithm

The canvas diagram auto-scales based on the input values. The origin point O is positioned with fixed top/left padding, and a scale factor (pixels per mm) is computed to ensure the full arm + sphere fits within the available canvas area, accounting for reserve space for labels, arrows, and dimension lines.

§ 9    **RESULT DISPLAY**

After clicking "Compute Solution", a result panel appears with an animated slide-up transition. It contains:

**Answer Boxes**

Two highlighted boxes displayed side by side in a grid:

| Box   | Label                      | Shows                                            |
|-------|----------------------------|--------------------------------------------------|
| Left  | Combined Moment of Weights | $M_{\text{weights}}$ in N·mm (clockwise about O) |
| Right | Required Biceps Tension T  | T in Newtons (for $\Sigma M_O = 0$ )             |

**Step-by-Step Working**

Below the answer boxes, a detailed step-by-step breakdown is displayed using monospace font with colour-coded values. Each step shows the formula, numeric substitution, and result. The steps are:

- 1. Compute weights** —  $W_{\text{arm}}$  and  $W_{\text{sphere}}$  from masses and gravity.
- 2. Geometry** — Calculate the forearm angle below horizontal ( $\alpha = 90^\circ - \theta$ ) and the horizontal distance from O to G.
- 3. Moment of forearm weight** —  $M_1 = W_{\text{arm}} \times x_G$ .
- 4. Moment of sphere weight** —  $M_2 = W_{\text{sphere}} \times d_A$ .
- 5. Combined moment** —  $M_{\text{total}} = M_1 + M_2$ .
- 6. Equilibrium** —  $T = M_{\text{total}} / d_T$ .

§ 10    **SIGN CONVENTION**

- **Positive x** → right
- **Positive y** → upward
- **Positive moment** → Counter-Clockwise (CCW)
- Weight forces act **downward** (negative y) — creating **clockwise** moments about O
- Biceps tension acts **upward** (positive y) — creating a **counter-clockwise** moment about O

## § 11 CANVAS INTERACTION DETAILS

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- All inputs have `input` event listeners that call `drawDiagram()` on every keystroke.
- The canvas uses `devicePixelRatio` scaling for crisp rendering on HiDPI displays.
- The canvas resizes responsively on window resize via a `ResizeObserver` / `resize` event.
- Force arrows, dimension lines, and labels are drawn programmatically — no bitmap assets required.

§ 12    **TECHNOLOGY DECISIONS**

| Decision       | Choice                            | Reason                                                             |
|----------------|-----------------------------------|--------------------------------------------------------------------|
| Architecture   | <b>Single HTML file</b>           | No build step; opens directly in any browser; zero dependencies    |
| Rendering      | <b>HTML5 Canvas 2D</b>            | Real-time redraws on every input; smooth rotations and scaling     |
| Styling        | <b>Embedded CSS</b>               | Dark theme with gold accent; uses Inter + DM Mono fonts            |
| Math           | <b>Built-in Math object</b>       | Sufficient for 2D moments; no external library needed              |
| Input handling | <b>parseField() with fallback</b> | Gracefully handles symbolic inputs while maintaining calculability |
| Diagram style  | <b>Anatomical rendering</b>       | Gradient fills, muscle bulges, 3D sphere for visual engagement     |

§ 13    **FUTURE ENHANCEMENTS**

- **Additional problems** — Extend to other Meriam & Kraige problems within the Moments & Couples chapter.
- **Home page** — Build a module selection screen linking to multiple problem solvers.
- **Export to PDF** — Use `window.print()` or jsPDF to export the diagram + solution as a PDF submission.
- **Save/load session** — Use `localStorage` to persist input values across sessions.
- **Animation mode** — Animate the moment construction with a parallelogram law visualisation.
- **3D problems** — Extend to 3D equilibrium using unit vectors (Meriam Ch. 3).
- **Friction problems** — Add impending motion, wedge friction, and belt friction solvers.

**REFERENCE TEXTBOOK**

**Engineering Mechanics: Statics**

J.L. Meriam & L.G. Kraige  
Wiley, 7th Edition (or later). Problem 2/50 appears in Chapter 2: Force Systems — Moments and Couples section.

END OF BLUEPRINT · PROBLEM 2/50 FOREARM MOMENT SOLVER

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